Signal Preprocessing and Artifact Removal for Brain-Computer Interfaces

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2022 IEEE 16th International Symposium on Applied Computational Intelligence and Informatics (SACI) | 978-1-6654-8125-0/22/$31.00 ©2022 IEEE | DOI: 10.1109/SACI55618.2022.9919582

***Abstract*—Brain-Computer Interfaces (BCIs) have become an important research field in the past decades. Interest has grown in this area given the tremendous potential of BCI systems of restoring, replacing, improving, enhancing and supplementing the activity of the central nervous system. Moving from laboratory refined circumstances to portable neuroprosthetics poses great challenges on signal processing and artifact removal. This paper discusses some methods used in signal preparation and preprocessing and also some techniques for artifact removal from electroencephalogram-recorded brain activity.**

Keywords—artifact removal, brain-computer interfaces, brain waves, electroencephalogram, feature extraction

I. Introduction

The term brain-computer interface (BCI) was first used by Jacques Vidal in the 1970s [1], [2] with the broad sense of any computer-based system that provides information on the function of the brain. Throughout the years, various reviews [3], [4], [5] have contributed to a more comprehensive and definitive definition of the BCI. Nowadays, it is considered that a BCI is a system that measures the activity of the central nervous system (CNS) and converts it into artificial output that modifies the interactions of the CNS with the internal or external environment by improving, restoring, enhancing, replacing, or supplementing the natural CNS output.

The activity of the CNS can be recorded in various ways using sensors placed within the brain, on the surface of the brain, or on the scalp. The brain activity produces both electric and magnetic signals. Noninvasively, at the scalp level, the electric signals are recorded primarily by electroencephalography (EEG) while the magnetic signals are recorded by magnetoencephalography (MEG). The former is the most extensively used method given that it is inexpensive, convenient, and is not confined to research settings, as MEG is due to the large dimensions and costs of the instrumentation. This paper will focus on signals acquired at scalp level, mostly by EEG.

Once the signals have been acquired, they go through a series of transformations, within a BCI system, before being converted into artificial outputs that act on the body or on the external environment. The brain signals are converted from analogue to digital, then they are digitally filtered, the next steps being feature extraction, feature translation, and the application of the outputs on the body or the environment [6].

This paper focuses on the signal preparation and preprocessing, the latter being the first stage of feature extraction which encompasses the removal of artifacts of both biological and nonbiological nature. The paper is organised as follows: Section II presents the signal preparation steps, Section III expounds the signal preprocessing steps, Section IV deals with artifact removal and Section V presents the conclusions.

II. Signal Preparation

From the brain signal recording to the extraction of features, the signal undergoes a couple of transformations that prepare it for further processing and digital storage. The signal acquired from the brain is of the order of μV to mV [7] and therefore it is usually amplified with the aid of a biosignal amplifier. The next step is to convert the signal from analog to digital or to digitize it. Most biosignal amplifiers also digitize the signal but if this is not the case, an analog-to-digital converter is needed. The conversion process consists of two steps: sampling and quantization.

During sampling, the voltage values of the continuous analog signal are captured periodically at a specific rate. The sampling rate is measured in samples/s or Hz. Choosing the best sampling rate is itself an interesting question because the higher the sampling rate the better the signal reconstruction but at the same time, the more digital space required to store the measured values. The Nyquist-Shannon sampling theorem [8] states that in order to achieve perfect signal reconstruction, the sampling rate has to be at least double the highest analog signal frequency (the Nyquist sampling rate). In the case of EEG recorded signals, there is no significant activity above 80 Hz [6] and therefore, a sampling rate of 160 Hz would facilitate a perfect reconstruction of the signal.

The quantization step converts each analog voltage sample into a binary value to be used and stored on the computer. Different quantization levels exist and they determine the number of bits used for representing the analog values and thus the number of possible discrete amplitude values. Given the discrete nature of the quantization levels, the analog voltages are rounded to the nearest amplitude value resulting in a quantization error [9]. Again, there is a trade-off between minimizing the quantization error by using higher number of bits and the digital storage space required.

Once the signal has been digitized, the fast Fourier transform algorithm [10] is used to convert the signal from time domain to frequency domain in order to simplify the mathematical analysis of the signal and also to enable the

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separation of the signal of interest from the contaminating signals. Once the signal is prepared, the extraction of the features can begin. The first stage of feature extraction is signal preprocessing and will be presented in the next section.

III. Signal Preprocessing

The signal preprocessing stage is also known as signal conditioning and is itself divided into multiple steps [6] which aim at enhancing the signal by eliminating the interferences of artifacts or by enhancing the signal characteristics that are of particular interest for the BCI application.

Frequency-range prefiltering is the first step in signal conditioning. Its aim is to keep only the signal that lies within the frequency range of interest for the particular situation. In this case, the brain activity of interest has a frequency of 0.5 to 40 Hz [11], [12], [13]. Studies have shown that the power of the EEG recorded signal decreases with 1/frequency, the signal is further attenuated by the scalp and skull and therefore the signal to noise ratio (SNR)

$$SNR = 10log_{10}\left(\frac{P_{signal}}{P_{noise}}\right) \quad (1)$$

of the EEG recorded signal above 40 Hz would be too low to be useful for the BCI application. P_{signal} in (1) is the signal power, and P_{noise} is the noise power. Narrower bandpass filters can be used to isolate specific frequency ranges that are of specific interest for various BCI applications such as the mu-band activity (8 – 12 Hz) and the beta-band activity (12 – 35 Hz) [7]. Independently of the BCI application, the bandpass filter width should be chosen with careful consideration in order to prevent the loss of information for both online and offline processing.

Once the frequency range of interest has been bandpass filtered, the data undergoes decimation and normalization. Decimation reduces the sample rate by removing the samples periodically [14]. Given that the signal has been filtered and the maximum frequency has been reduced to 40 Hz, as discussed in Section II, the signal can be downsampled to the Nyquist frequency of 80 Hz. Signal normalization is a technique that eliminates the amplitude differences by subtracting from each signal its mean value and dividing the result by the signal's variance. It is useful when comparing EEG signals recorded from different brain regions and the signals vary largely in amplitudes but the signal dynamics, which might be the ones of interest for the BCI application, are comparable after normalization [15]. In some BCI applications, where the amplitude differences between the signals are of interest, normalization is not used.

When it comes to EEG as a recording method, the values measured are in fact electrical potential differences between two electrodes, measured in volts (subunits), the voltage obtained is referred to as a channel. Usually, all channels have a common electrode, the reference, that is placed in a region free of brain activity such as the mastoids, the ears or the tip of the nose [7]. Given that each electrode records the activity of many proximal brain sources and the sensitivity of the channel depends on many factors like the size, orientation, or location of the brain sources, the spatial filtering technique, which is the third step in signal conditioning, enables the increasing of the signal-to-noise ratio by making weighted combinations of channels in order to make them more sensitive to specific brain sources. Spatial filters are represented as the following matrix multiplication:

$$\begin{bmatrix} w_{11} & w_{12} & \cdots & w_{1n} \\ w_{21} & w_{22} & & \\ \vdots & & \ddots & \\ w_{m1} & & & w_{mn} \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & & \\ \vdots & & \ddots & \\ x_{n1} & & & x_{np} \end{bmatrix} = \begin{bmatrix} y_{11} & y_{12} & \cdots & y_{1p} \\ y_{21} & y_{22} & & \\ \vdots & & \ddots & \\ y_{m1} & & & y_{mp} \end{bmatrix} \quad (2)$$

of the digital signal samples (x_{np}) with the channel weights (w_{mn}), resulting in the spatially filtered channels (y_{mp}), where there are n channels, m weights for each channel and p consecutive signal samples.

There are multiple approaches for choosing the spatial filter weights, this can be done in a data-dependent or data-independent manner. One of the most common data-independent spatial filters is the common average reference (CAR), which reduces the impact of the artifacts that are common across all electrodes by recording the channels with a common reference, computing the global mean at each time point and subtracting it from each individual channel [16]. A typical formula for this filter is [16]

$$P_i = \Delta P_{iR} - \frac{1}{n}\sum_{j=1}^{n} \Delta P_{jR} \quad (3)$$

where P_i is the potential at electrode i, ΔP_{iR} is the potential difference between electrode i and the reference (highlighted using the subscript R) and n is the total number of electrodes in the montage. McFarland et al. [17] showed that proper spatial filtering of EEG data helps maximizing the SNR and thus improves the speed and accuracy of EEG-based BCI applications. On the other hand, data-dependent spatial filters are derived from each subject's BCI data. Even though they are more tedious to implement, they might provide better results in certain situations, especially when not much is known about the characteristics of the targeted brain activity. Principal component analysis (PCA) [18] is one of the most widely used methods for deriving data-dependent spatial filters. This method determines a spatial filter that multiplied with the original set of signals results in a new set of signals with the same dimensionality but with the channels being uncorrelated with each other. The new signals (principal components) are linear combinations of the original signals sorted by the amplitude variance. In other words, PCA enhances certain brain activity, especially when it is correlated with the conditions of the BCI task. One of the PCA's particularities is that most of the amplitude variance is captured by the first few components and therefore the dimensionality of the problem can be reduced by removing the other components. However, the removal of the lower components should be done carefully given that this method is not able to separate certain brain activity from artifacts particularly when they have similar amplitudes [19].

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The last step in the signal conditioning stage is the removal of biological and non-biological artifacts. As this is an important part, it will be discussed in detail in Section IV.

IV. Artifact Removal

Detecting and removing artifacts from the recorded brain activity is a crucial step in achieving accurate results. Given the very good insulation provided by the skull and scalp and the low voltage of the brain activity, the recordings are very easily contaminated by artifacts attributable to both biological and non-biological sources.

Environmental interference is an artifact of non-biological nature. It comprises interferences from power lines and other electric devices that create electrical fields in the body detectable by EEG electrodes. This type of interference is easy to detect given that it is a continuous sinusoidal signal at 50 or 60 Hz (50 Hz in Europe) that can be removed using a bandpass frequency filter with a band of 45-55 Hz or, given that the interference affects all the electrodes (channels), a CAR spatial filter can be used to remove it. Furthermore, interference may be produced by faulty electrodes or by high-impedance electrodes. There are various automatic approaches for detecting and removing these kind of artifacts [20], [21] but Zou et al. [22] have developed an algorithm that is suitable for portable EEG systems with lower electrode density, which are the kind of systems used nowadays for BCI applications. The algorithm consists of EEG data preprocessing, independent component analysis (ICA) and identification of artifact related independent components (ICs). During preprocessing the EEG data is bandpass filtered and split into epochs. ICA is a well-known statistical approach that aims at finding a separation matrix W such that

$$s = W\ x\ v, \tag{4}$$

where s is an array (or vector) of unknown statistically independent brain and artifact sources and v is an array of recordings. ICA is an efficient method because the resulting ICs are both brain and non-brain sources. In order to separate the artifacts from the brain-related ICs, several types of features are extracted. Artifacts like eye blinks cause the amplitude of the IC to raise abruptly. This fluctuation can be captured using kurtosis [23] which is a measure of relative peaks in the amplitude distribution. Given that the artifact and the brain ICs are projected on different electrode groups, the median of each IC's topography weight is utilized to separate the artifacts

$$f_{spatial} = median(tp(1)_i, tp(2)_i, \dots\ tp(n)_i), \tag{5}$$

where $tp(n)$ is the IC's topography weight matrix W^{-1} for electrode n. As discussed above, the power of most brain signals falls into the 1-40 Hz range and the artifacts have dissimilar power distributions (20-50 Hz for muscular activity). Therefore, the average band power of delta, theta, alpha, beta and gamma bands help highlighting the differences between artifacts and brain activity in terms of the row matrix

$$f_{spectral} = [F(delta)\ F(theta)\ F(alpha)\ F(beta)\ F(gamma)], \tag{6}$$

where the average band power $F()$ can be calculated using MATLAB. Another difference between brain related activity and artifacts is that the epochs containing artifacts have no pattern and are not similar to one another while the ones containing brain activity exhibit certain similarities. The correlation value of such epochs is used to measure similarity

$$f_{similarity} = \frac{\sum_{m=1, m\neq i}^{N} corr_{s_i s_m}}{N-1}, \tag{7}$$

where $corr_{s_i s_m}$ is the correlation coefficient between the ICs i and m and N is the total number of epochs. For each IC, a feature vector with 8 dimensions (1 temporal, 1 spatial, 5 spectral, and 1 similarity feature) is extracted and then used in a hierarchical clustering algorithm, which automatically distinguishes between the artifact and brain activity related ICs.

Biological artifacts arise from the electrical activity generated by muscle contractions, eye blinks, eye movements, or heartbeats. The last three categories are easier to identify given their periodic nature. Eye blinks usually create a high amplitude signal, greater than the brain signals of interest that corrupts data all across the recording site, even at the back of the head. In order to mitigate this problem, pairs of electrodes are placed around the eyes to measure the electrooculogram. Algorithms based on blind source separation [24] are available for using the electrooculogram data to remove the eye blink artifacts from the EEG recorded signal. Eye movement artifacts have an even stronger effect than the eye blink artifacts and they usually occur at short periods after the blinks. Spatial filtering techniques based on weights derived from a regression with respect to the electrodes placed around the eyes are useful for removing this kind of artifacts [6]. Heartbeat artifacts contaminate the signal when an electrode is placed in the proximity of a blood vessel that induces voltage changes by its contraction and dilation. PCA and ICA can be used for removing pulse artifacts but a better approach is average artifact subtraction [25] that computes an artifact template for each occurrence and then averages the templates and subtracts them from the data.

Across time, many algorithms based on various techniques have been developed to remove biological artifacts [26] but one of the most challenging problems remains the removal of muscle activity artifacts. Muscle activity is not periodic, has a wide frequency range and can affect the entire range of electrodes. It is usually more prominent in frontal, occipital and temporal regions but can as well contaminate central regions. Goncharova et al. [27] have investigated this problem more in depth and reached the conclusion that muscle activity is cross spectral and can even resemble beta peaks, which is the frequency we are most interested in for BCI applications. Therefore, frequency filtering would not be effective and removal of such artifacts would rather require recording from peripheral scalp locations.

V. Conclusion

This paper presented the signal preparation and conditioning stages from a BCI system. It focused on the transformations the EEG recorded signal undergoes before features can be extracted and also detailed some methods that

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are used in order to achieve this. Furthermore, the subject of artifact removal has been discussed together with methods and techniques for removing the contaminations from the signal.

Further research will focus on feature extraction itself, feature translation and various machine learning approaches for converting the features into the desired output able to act on the body or on the environment and fulfil the BCI's objective.

ACKNOWLEDGMENT

This work was supported by a grant of the Romanian Ministry of Education and Research, CNCS - UEFISCDI, project number PN-III-P4-ID-PCE-2020-0269, within PNCDI III.

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